The Collateral Impact of the Three-Dimensional Jacobian Counterexample: the Cascade of Implications

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Abstract

On 2026-07-20, Levent Alpöge announced an explicit counterexample to the Jacobian conjecture in dimension 3, found working with the language model Claude Fable 5 (Anthropic). The credit for every consequence documented here belongs to that single map. This paper documents what else that map decides: through implication chains published over five decades, each, except Mathieu's, located in its primary source (EXP-016, EXP-018, EXP-136; Mathieu's 1997 paper was not located, and that link rests on Zwart's review of it), the counterexample resolves or partially resolves the Mathieu conjecture, the Dixmier conjecture, the Poisson conjecture, the Gaussian moments conjecture, Zhao's vanishing conjecture, the Image conjecture, and the symmetric (gradient) form of the Jacobian conjecture. We state each corollary with its chain and its reference, list precisely what remains open, and describe the verification method, which turns a one-line certificate into a status map of a whole neighborhood of problems. The symmetric/gradient and vanishing corollaries are made explicit: a non-invertible gradient Keller map $\mathrm{id}+\nabla f$ on $\mathbb{C}^{48}$, with f a homogeneous Hessian-nilpotent quartic (382 monomials over $\mathbb{Q}(i)$) and an exact two-point collision, built from Thompson's dimension-24 cubic-homogeneous form by the de Bondt–van den Essen construction and verified end to end in exact arithmetic. A homogeneous form of the counterexample in dimension 15, built by de Bondt and verified here exactly, shows that the Mathieu conjecture fails for $SU(N)$ for every $N \geq 15$.

1 The counterexample and its verification

The counterexample is the polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$, $u = 1 + xy$,

$$F = \big(u^3 z + y^2 u(4 + 3xy),\ \ y + 3xu^2 z + 3xy^2(4 + 3xy),\ \ 2x - 3x^2 y - x^3 z\big),$$

with $\det JF = -2$ everywhere and $F(0,0,-\frac{1}{4}) = F(1,-\frac{3}{2},\frac{13}{2}) = F(-1,\frac{3}{2},\frac{13}{2})$ [1]. We validated it independently in exact rational arithmetic and computed the full fiber over the repeated value: it consists of exactly the three announced points (EXP-001). Every consequence below is a corollary of this validation plus one or more published implications; for each implication except Mathieu's the primary source was located and its direction and dimension index confirmed (EXP-016, EXP-018 and EXP-136); Mathieu's 1997 paper was not located, and that link rests on Zwart's review. The corollaries of Section 2 add no mathematics beyond this verification and the composition of the published implications, except the statement that F is not affinely of homogeneous type, checked by a Gröbner basis computation (EXP-136); the explicit witness of Section 4 is a construction from published ingredients (Thompson's cubic-homogeneous form and the de Bondt–van den Essen reduction).

2 The cascade

Corollary 1 (Smale's 16th problem)**.** *The Jacobian conjecture is false for every $N \geq 3$; the two-variable case remains open.*

Corollary 2 (Mathieu conjecture, 1997)**.** *Mathieu proved that his conjecture, taken for all compact connected Lie groups, implies the Jacobian conjecture [2] (the original paper was not located; the implication is checked in Zwart's review, [6, Theorems 4.16 and 4.23]); Duistermaat and van der Kallen proved the abelian case [5]. At a fixed N the argument shows that the conjecture for $SU(N)$ makes every Keller map $x - h$ of $\mathbb{C}^N$ with h homogeneous invertible [6, Theorems 4.16 and 4.23]. De Bondt's homogenization of the counterexample [18] gives such a map on $\mathbb{C}^{15}$, of degree 7, which is not injective (Section 3; the cubic-homogeneous form on $\mathbb{C}^{24}$ of Section 4 is another), so the Mathieu conjecture is* ***false for*** *$SU(N)$, $N \geq 15$, after adjoining identity coordinates for $N > 15$. For $SU(2)$ an explicit counterexample is given in [3]. For $3 \leq N \leq 14$ the argument does not reach F: its components have total degrees $7, 6, 4$ and its second partial derivatives have no common zero, so F is not affinely equivalent to a map $x - h$ with h homogeneous (EXP-136), and the reduction of F to that form adds variables [4].*

Corollary 3 (Dixmier conjecture, 1968)**.** *$JC(2n)$ implies* Dixmier(n)*, and* Dixmier(n) *implies $JC(n)$ [7, 8]. Since $JC(n)$ is false for every $n \geq 3$, the second implication gives that* Dixmier(n) *is false for every $n \geq 3$: the* ***full Dixmier conjecture is false****. Ranks 1 (Dixmier's original question) and 2 remain open; the first implication gives no information on rank 2, because it runs from $JC(4)$ to* Dixmier(2)*. A proof of rank 1 is claimed in [9].*

Corollary 4 (Poisson conjecture, 2007)**.** *For each n, $JC(2n) \Rightarrow$* Poisson$(n) \Rightarrow$ Dixmier$(n) \Rightarrow$ *$JC(n)$ in any characteristic, where* Poisson(n) *concerns the canonical Poisson algebra in $2n$ variables [10, Theorem 7]. Hence* Poisson(n) *is false for every $n \geq 3$: the* ***full Poisson conjecture is false****. Index 2 is settled separately: Long gives explicit $R, T, D, S \in \mathbb{Q}[x, q, p, z]$ with $\{D, R\} = \{S, T\} = 1$ and the other brackets zero, where $R = x(2 - 3xq)$ is reducible, so $(x, q, p, z) \mapsto (R, T, D, S)$ is a Poisson endomorphism of the canonical Poisson algebra in four variables that is not an automorphism [12]. Hence* Poisson(n) *is false for every $n \geq 2$. Index 1 remains open; a false* Poisson(2) *gives no information on* Dixmier(2)*, since the implication runs from* Poisson(2) *to* Dixmier(2)*.*

Corollary 5 (Gaussian moments conjecture)**.** *GMC implies JC [11]; hence* ***GMC is false*** *at some finite dimension.*

An explicit counterexample in three variables, with P of five terms and total degree 4, is given in [13]; GMC is therefore false in every dimension $n \geq 3$. GMC holds in dimension 1 [11, Proposition 4.2], and a proof of the two-variable case is claimed in [14].

Corollary 6 (Zhao's vanishing conjecture)**.** *The vanishing conjecture for Hessian-nilpotent polynomials is equivalent to JC [15, 16]; hence it is* ***false****: there exists a Hessian-nilpotent quartic P with $\Delta^m P^m = 0$ for all m but $\Delta^m P^{m+1} \neq 0$ for infinitely many m.* An explicit witness is given in Section 4.

Corollary 7 (Image conjecture)**.** *The Image conjecture implies the vanishing conjecture [17]; hence it is* ***false in some dimension****.*

Corollary 8 (Symmetric / gradient Jacobian conjecture)**.** *JC reduces to cubic homogeneous maps with symmetric Jacobian [19]; with the Zhao equivalence this yields a dimension in which a* ***gradient Keller map is non-invertible****.* An explicit such map, in dimension 48, is given in Section 4. *Smaller failing dimensions for the vanishing conjecture,* 28 *for a non-homogeneous and* 30 *for a homogeneous potential over the Gaussian integers, are reported in [18]; they are not re-verified here.*

3 A homogeneous witness in dimension 15 (EXP-138)

De Bondt [18] starts from the volume-normalized form $F = (x_1 - 3x_1^2x_2 - x_1^3x_3,\ 3v^2x_1x_3 + (9v^2 - 6v - 2)x_2,\ v^3x_3 + (6v^2 + 2v)x_2^2)$, $v = 1 + 2x_1x_2$, of the counterexample, with $\det JF = 1$ and $F(0, 1, -8) = F(-1, 1, 4)$, and reduces the degrees of its components with new variables until every component of the homogenized map has the form $x_i + H_i$ with H_i homogeneous of degree 7. We re-implemented this construction independently and verified the output exactly: $G = x + H$ on $\mathbb{C}^{15}$, $(JH)^{14} = 0$, so $\det JG = 1$, and the distinct integer points $(0, 1, -8, 0, \ldots, 0, 1)$ and $(-1, 1, 4, -4, -1, 48, 12, -36, -12, 32, 8, -40, -12, 12, 1)$ have the same image. With $h = -H$ this is a non-invertible Keller map $x - h$ with h homogeneous, which is what the Mathieu corollary needs.

4 The explicit witness in dimension 48 (EXP-041)

As of 2026-07-22 no explicit symmetric, gradient, or Hessian-nilpotent witness had been published (on a search of the literature; the July 2026 consequence notes are non-constructive in dimension). We construct one in three steps, each verified in exact arithmetic.

Input. W. Thompson's explicit Bass–Connell–Wright cubic-homogeneous form of the counterexample [20]: $G = U + H$ on $\mathbb{Q}^{24}$, H cubic homogeneous with 54 monomials, with a published rational collision pair. We verified this form independently (homogeneity; the exact collision; nilpotency of JH by an exact sparse chain). The nilpotency index is 18, not 17 as stated there: $(JH)^{18} = 0$ while $(JH)^{17}$ retains six terms.

Construction (de Bondt–van den Essen [19]). The potential $f = -i \sum_{j=1}^{24} H_j(x + iy)\, y_j$ is a homogeneous quartic in 48 variables (382 monomials over $\mathbb{Q}(i)$) with nilpotent Hessian (their characteristic-polynomial identity reduces this to the nilpotency of JH). The triangular

conjugation $S^{-1} \circ (\mathrm{id} + \nabla f) \circ S = (x + H(x),\ y + JH(x)^t y - iH(x))$ was verified identically in all 48 components, so $\widetilde{F} = \mathrm{id} + \nabla f$ is a symmetric Keller map ($\det J\widetilde{F} = 1$).

The collision. With (u, v) Thompson's collision pair and $w = i\,(I + JH(v)^t)^{-1}(u - v)$, the points $p_1 = (u, 0)$ and $p_2 = (v - iw,\ w)$ are distinct and satisfy $\widetilde{F}(p_1) = \widetilde{F}(p_2)$ exactly over $\mathbb{Q}(i)$. **Hence the Hessian conjecture fails in dimension 48 with an explicit witness**, and $P_\star = -f$ is an explicit homogeneous Hessian-nilpotent quartic falsifying the vanishing conjecture (the finite certificate of failure is the collision itself; the consistency check $\Delta P_\star^2 \neq 0$, forced by Zhao's own Corollary 3.9 for any failing potential, holds with 9028 terms). The reality obstruction is intrinsic: a nonzero real Hessian-nilpotent polynomial does not exist, so the $\mathbb{Q}(i)$ coefficients are unavoidable. Symmetric positive results (dimension ≤ 4 nonhomogeneous, ≤ 5 homogeneous, de Bondt–van den Essen and van den Essen–Washburn) leave the minimal witness dimension open between those bounds and 48.

5 What remains open

The two-variable Jacobian conjecture; Dixmier at ranks 1 and 2 (a proof of rank 1 is claimed in [9]) and Poisson at index 1; Mathieu for $SU(N)$ with $3 \leq N \leq 14$ and for the other non-abelian compact groups; the moments statement in dimension 2 (a proof is claimed in [14]); the minimal failing dimensions of the vanishing and Image statements (for the symmetric/vanishing pair the witness of Section 4 pins the failing dimension at ≤ 48 by an independently verified example and [18] reports ≤ 28, with the known positive results leaving the minimum open above 5); and explicit witnesses for the remaining non-constructive rows. Not included in this list: the Markus–Yamabe conjecture (refuted independently in 1997, not a corollary of the counterexample) and statements whose implication direction does not propagate the falsity of JC.

6 Method and data availability

All computations use exact arithmetic (no floating point) and are archived in the repository `https://github.com/fsantibanezleal/CAOS_RESEARCH`. An identifier EXP-*nnn* denotes one archived record in the experiment folders of the problem, `problems/algebraic-geometry/jacobian-conjecture/experiments/`: a computation with its script and output, or a check of published implications against their primary sources; each record states its question, method and result. The records used here are EXP-001 (exact validation, full fiber), EXP-016 and EXP-018 (each implication except Mathieu's located in its primary source and its direction confirmed), EXP-041 (the dimension-48 witness, with its input data, script and outputs) and EXP-136 (the dimension index of each implication; the homogeneous-type test of F; exact checks of the explicit counterexamples of [13, 3] for the first powers), and EXP-138 (the independent re-implementation and exact verification of the dimension-15 homogeneous witness).

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